🛡️ ISEC2700 – Information Security

MP1 Foundations: Identifying Security Issues in a Small Business Environment

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1. Conceptual Explanation

Before security professionals can protect an organization, they must first **understand what can go wrong**.

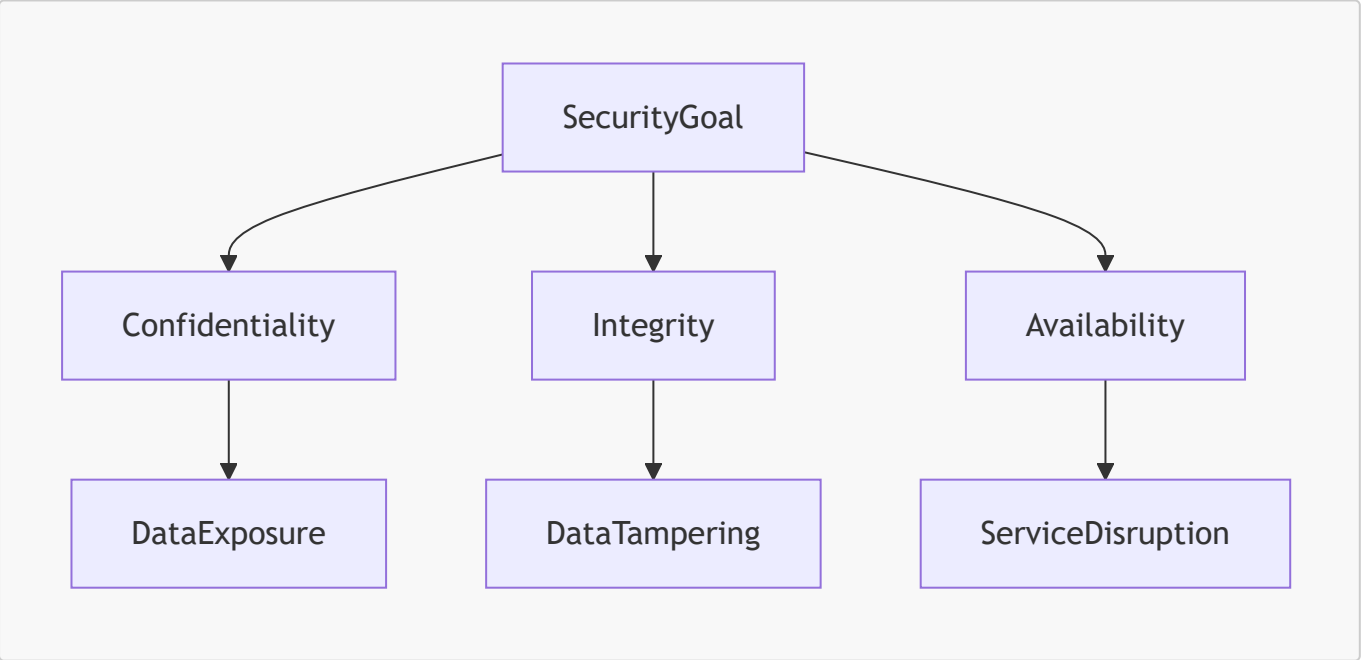
This process is called **security risk identification**.

Security issues do not exist in isolation. They exist because:

- systems store **valuable assets**
- attackers attempt to exploit **vulnerabilities**
- threats can cause **damage to business operations**

The goal of security analysis is to systematically identify weaknesses that could lead to a **confidentiality, integrity, or availability failure**.

This concept is grounded in the **CIA Triad**.



Understanding CIA allows analysts to answer the critical question:

> *What security property is at risk if this system fails?*

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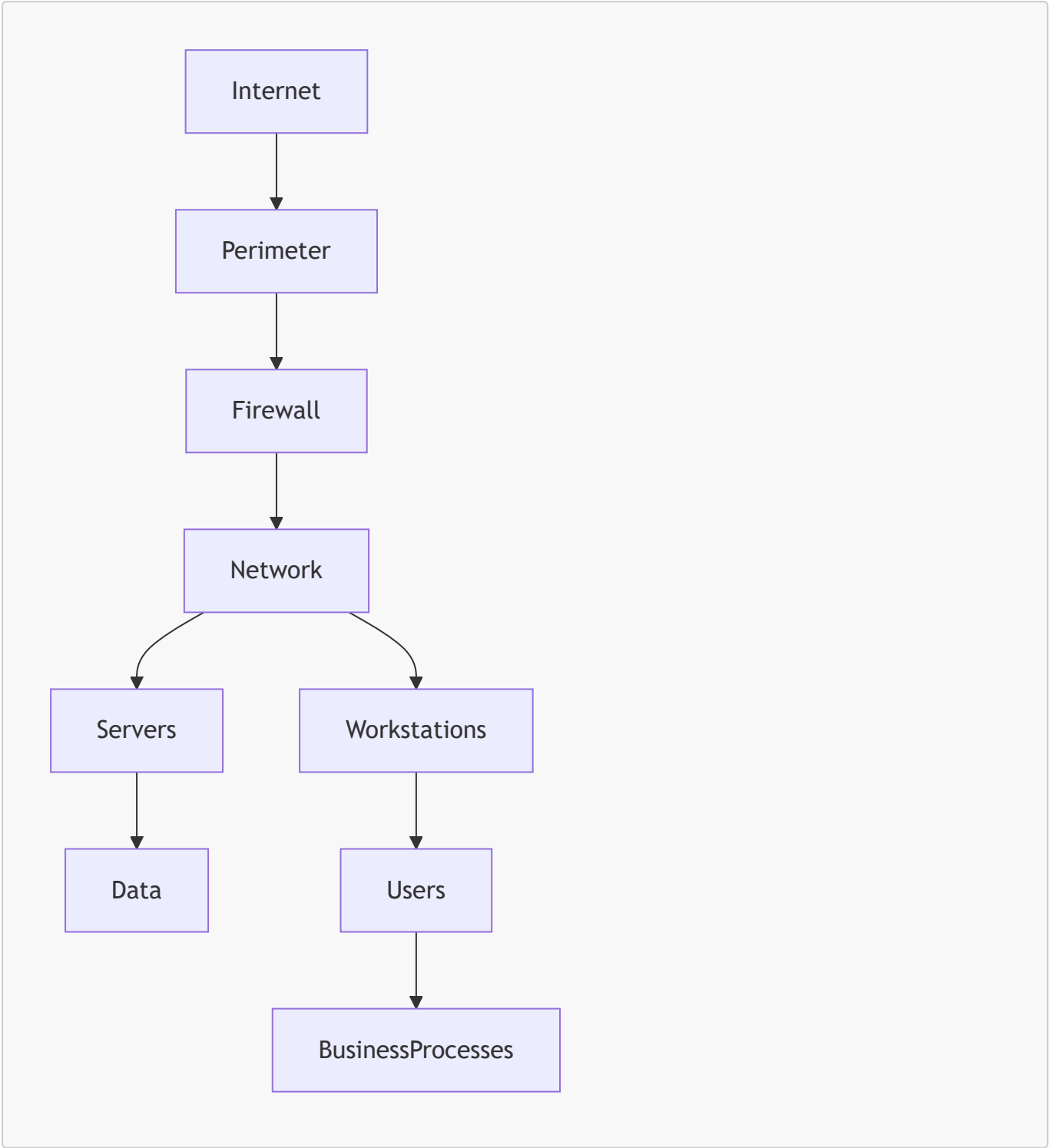
2. Architecture-Level Context

(How real enterprise environments are structured)

In professional environments, security issues are identified by analyzing **system architecture layers**.

These layers exist in nearly every enterprise network.



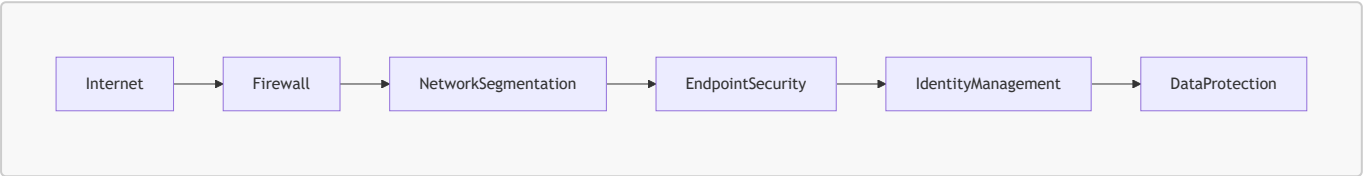
Each layer introduces **different types of security risks**.

Architecture Layer	Typical Security Issues
Perimeter	Firewall misconfiguration
Network	Flat networks / no segmentation

Architecture Layer	Typical Security Issues
Systems	Unpatched servers
Identity	Weak authentication
Data	Improper access control

Security analysts therefore evaluate systems **layer by layer**.

This is known as **defense-in-depth analysis**.



The goal is that **if one control fails, another still protects the organization**.

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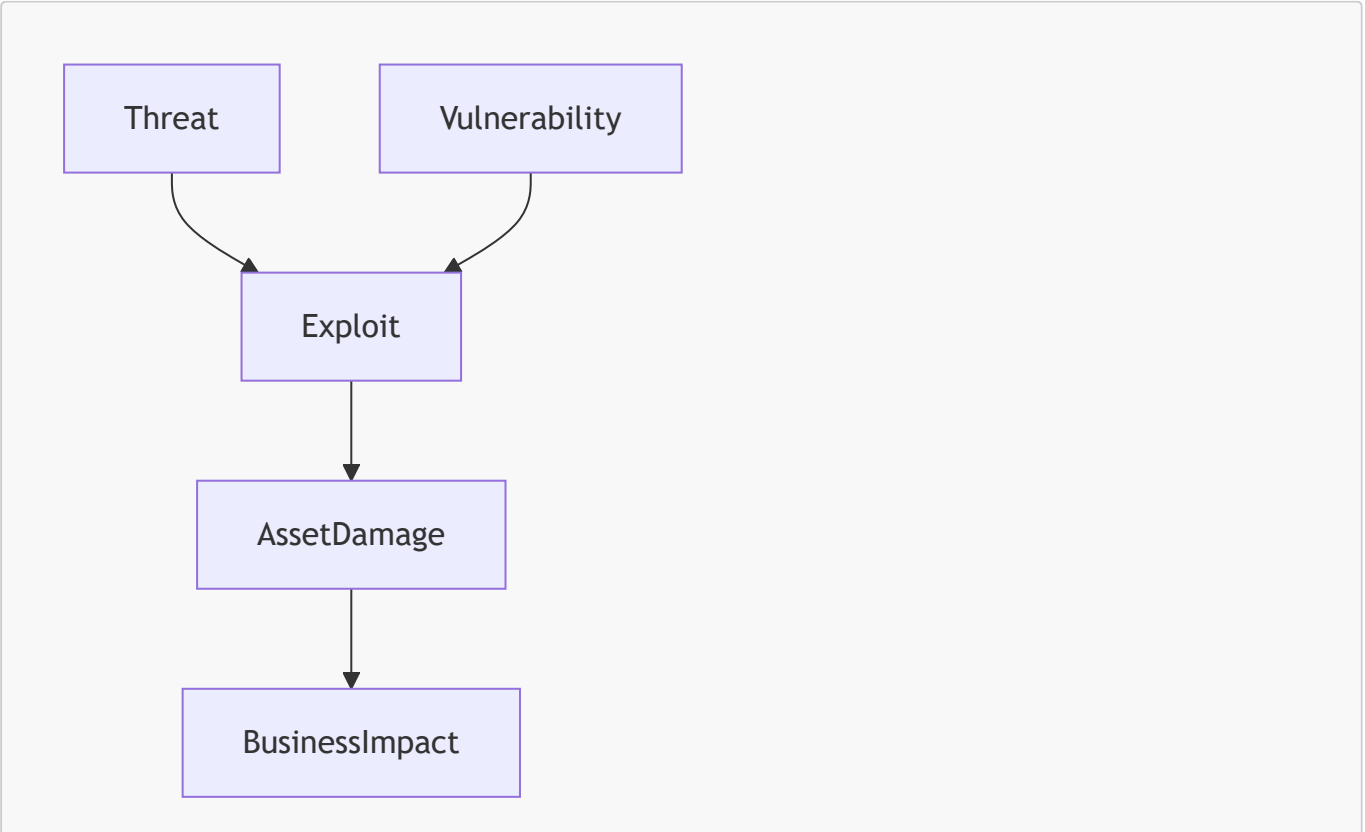
3. Understanding Risk in Cybersecurity

Before identifying issues, students must understand the relationship between:

- **Threats**
- **Vulnerabilities**
- **Assets**
- **Impact**

These components form the basis of **risk analysis**.



Definitions

Component	Meaning
Asset	Something valuable
Threat	Potential attacker or event
Vulnerability	Weakness that can be exploited
Impact	Damage to the organization

Risk is commonly described as:

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Risk = Threat × Vulnerability × Impact
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Security professionals must **identify vulnerabilities before attackers do**.

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4. Risk Analysis vs Risk Assessment

Students often confuse these two terms.

They represent **different phases of the security evaluation process**.

Risk Analysis

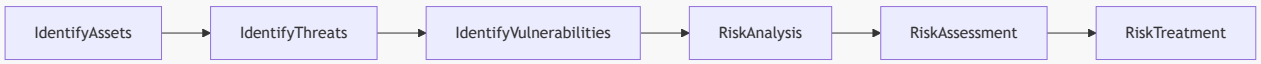
Risk Analysis determines:

- how likely an attack is
- how severe the impact could be

Risk Assessment

Risk Assessment determines:

- how serious the risk is overall
- which risks must be addressed first



In professional environments, this process is formalized in frameworks such as **ISO 27005** and **NIST Risk Management Framework**.

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5. Step-by-Step Operational Security Analysis

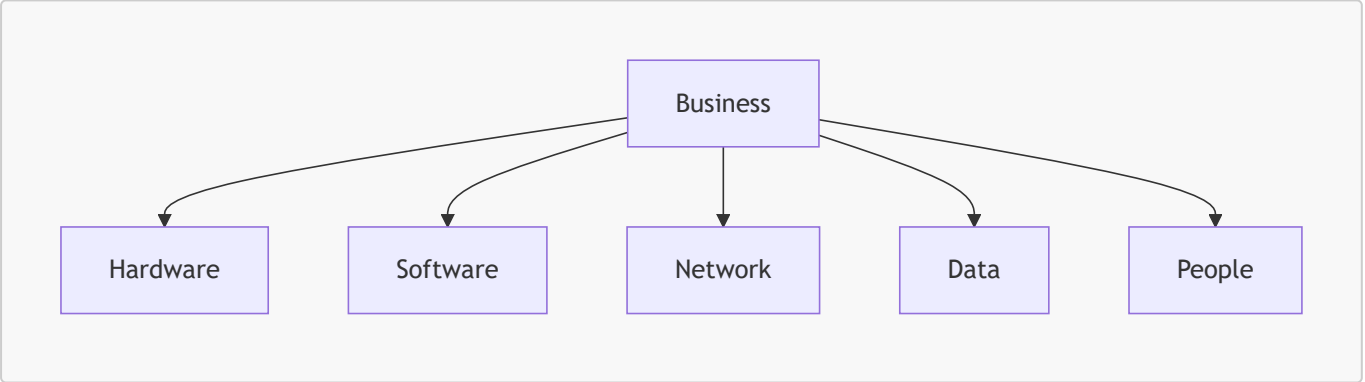
Security analysts follow a systematic methodology.

Step 1 – Identify Assets

Every security investigation begins with understanding **what needs protection**.

Typical assets include:

Asset Type	Examples
Hardware	Servers, laptops
Software	Business applications
Network	Routers, switches
Data	Customer records
Human	Employees



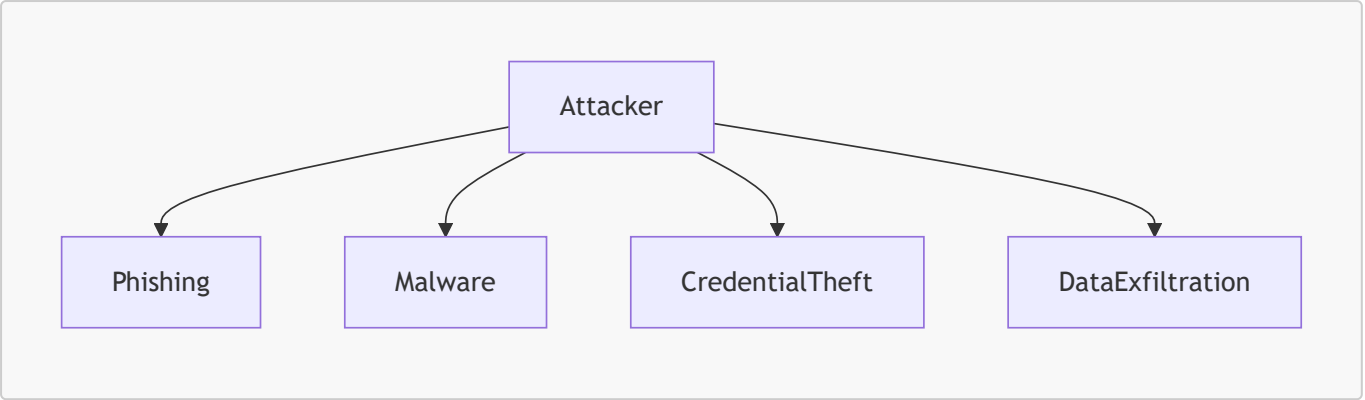
Without understanding assets, security analysis becomes guesswork.

Step 2 – Identify Threats

Threats are events that could damage assets.

Examples include:

- ransomware
- phishing attacks
- insider misuse
- hardware theft



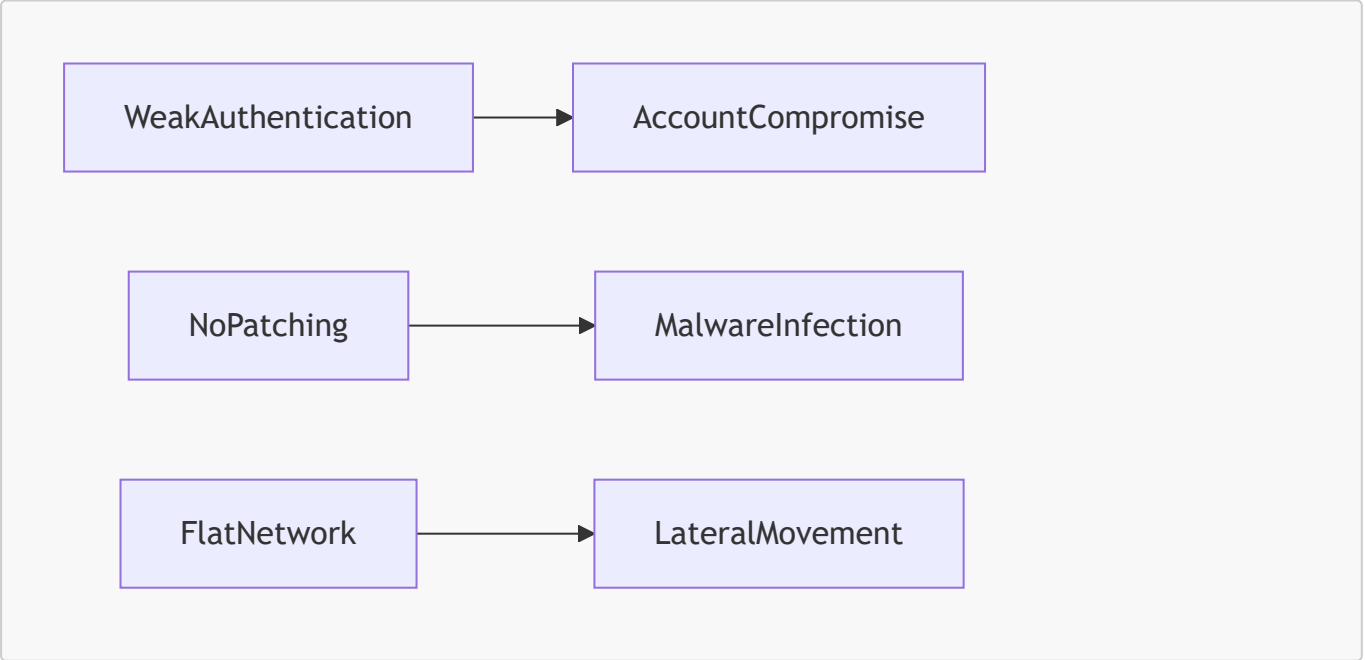
Threat identification helps analysts understand **what attackers are trying to accomplish**.

Step 3 – Identify Vulnerabilities

A vulnerability is a weakness that makes an attack possible.

Examples:

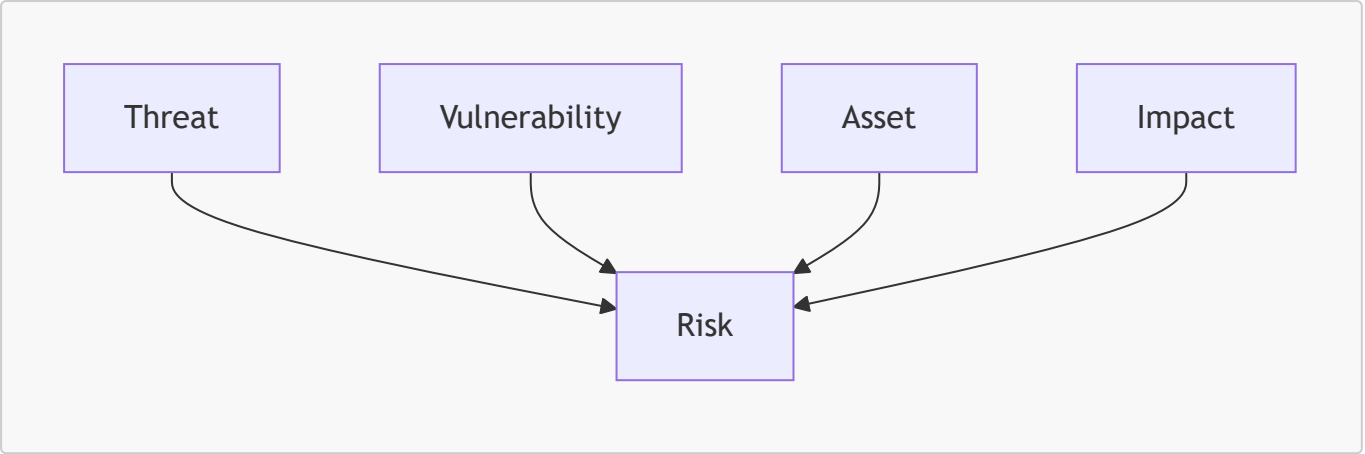
Vulnerability	Result
No MFA	Account takeover
Unpatched systems	Malware infection
Shared passwords	Lack of accountability
Flat network	Lateral movement



Security analysts look for **misconfigurations, missing controls, and outdated systems**.

Step 4 – Risk Analysis

Risk analysis evaluates **likelihood and impact**.

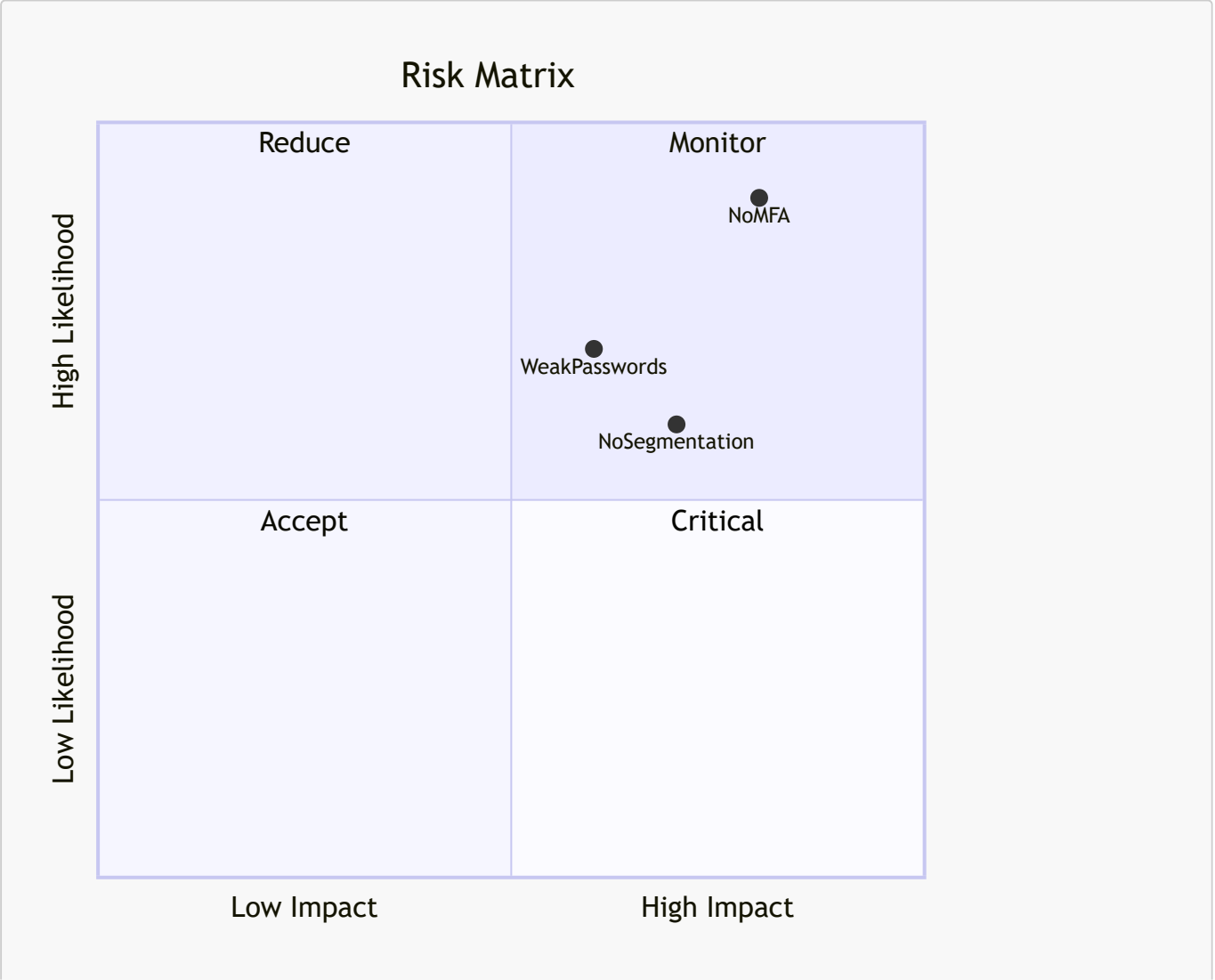


Example:

Risk	Likelihood	Impact
Ransomware	High	High
Data leak	Medium	High

Step 5 – Risk Assessment

Risk assessment determines **priority levels**.



The most critical risks must be addressed first.

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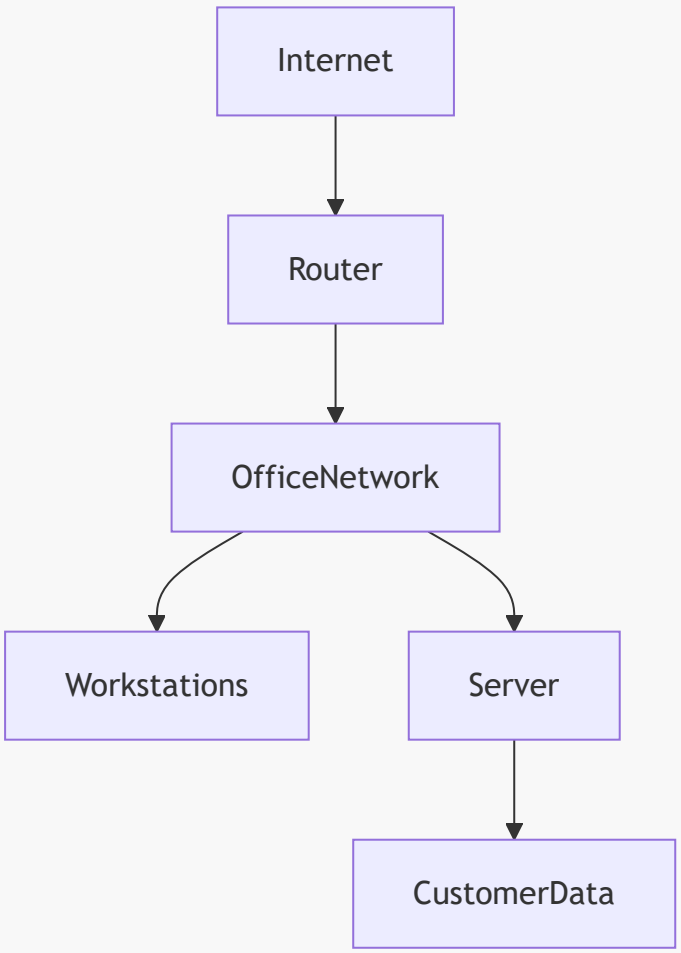
6. Real-World Small Business Scenario

Consider the following environment.

MapleTech Accounting

- 22 employees
- Windows Server (Active Directory)
- Microsoft 365 (no MFA)
- Shared Wi-Fi password
- Consumer router
- USB backups stored on-site

This environment contains multiple potential security issues.



Students must analyze the environment and identify weaknesses.

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7. Real-World Failure Cases

Understanding real incidents helps explain why these issues matter.

Case 1 – Ransomware Attack

Cause:

- Unpatched systems
- No endpoint protection

Outcome:

- File server encrypted
- Business operations halted
- Company forced to pay ransom

Case 2 – Business Email Compromise

Cause:

- No MFA on Microsoft 365

Outcome:

- attacker sends fraudulent invoices
- financial loss occurs

Case 3 – Insider Data Theft

Cause:

- excessive user permissions
- no monitoring

Outcome:

- employee copies customer database

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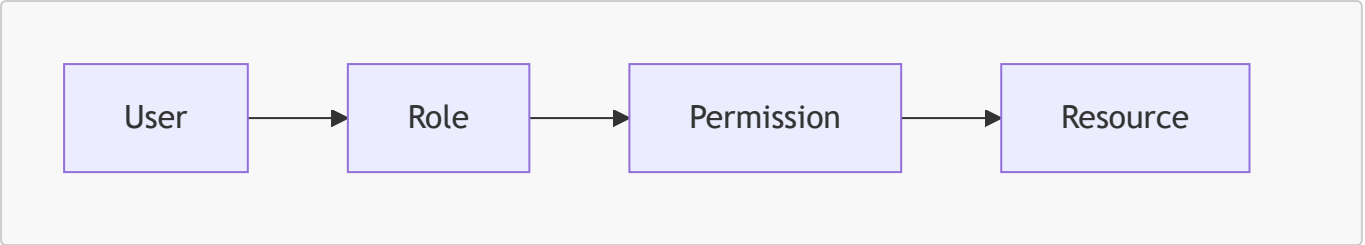
8. Best Practices and Security Principles

Security professionals follow several guiding principles.

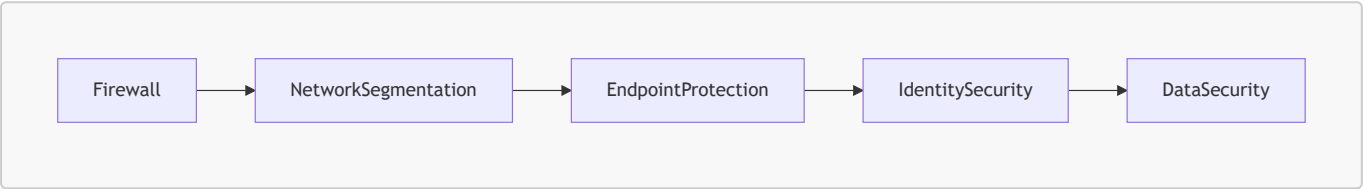
Least Privilege

Users should only have access to what they need.



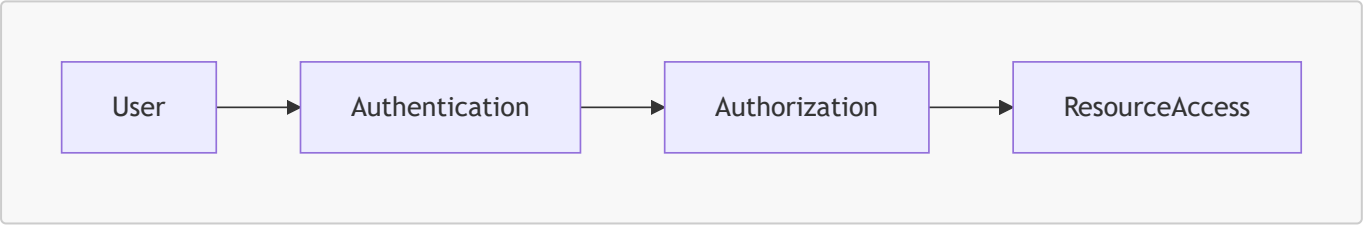
Defense in Depth

Multiple security layers protect systems.



Zero Trust Philosophy

Trust nothing without verification.



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9. Why This Matters for MP1

The purpose of MP1 is **not simply to list vulnerabilities**.

Students must demonstrate that they can:

1. Understand system architecture
2. Identify assets and threats
3. Recognize vulnerabilities
4. Analyze risk logically
5. Prioritize security improvements

This reflects **real cybersecurity analysis workflows used in industry**.

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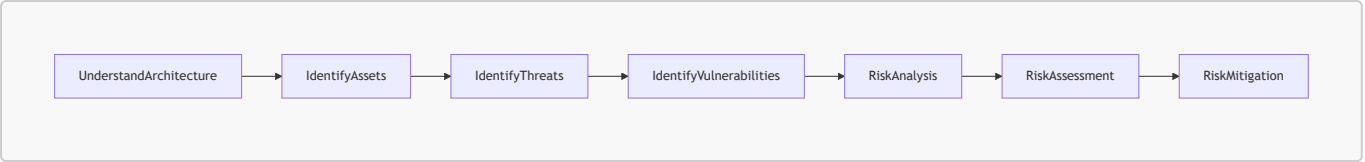
10. Key Takeaway

Security professionals do not randomly search for problems.

They follow a structured investigation process:



This methodology allows security analysts to identify weaknesses **before attackers exploit them**.